

Flexible Frequentist Nonlinear Model Selection

A Genetically Modified Feature Search with the Smooth Information Criterion

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Abstract

We introduce the *Flexible Frequentist Model Selection* (FFMS) framework, a frequentist reformulation of the Bayesian Genetically Modified Nonlinear Model Configuration of Hubin et al. [2021]. The central statistical engine replaces Bayesian marginal likelihood computation via Markov chain Monte Carlo with a continuous relaxation of the ℓ_0 sparsity penalty—the *Smooth Information Criterion* (SIC)—minimised by quasi-Newton optimisation with an ε -telescope schedule. The feature-specific SIC penalties are calibrated to the Bayesian information criterion (BIC), assigning each symbolic feature a cost proportional to its tree-depth complexity $oc_j \log n$, in exact correspondence with the log-prior penalties inherited from the Bayesian parent framework. The resulting inclusion probabilities drive a genetic population-transition kernel that breeds new candidate features through multiplication, nonlinear modification, and projection operations, mirroring the inter-population moves of Hubin et al. [2021] while remaining entirely free of prior specification. We prove the population-consistency of the SIC penalty and demonstrate, via a benchmark experiment on exoplanetary data, that FFMS rediscovers Kepler’s Third Law with a SIC value of -2900 , outperforming the reference model that encodes the known physical law directly ($\text{BIC} = -2749$). An open-source R implementation is provided at <https://github.com/aliaksah/FFMS>.

Keywords: automatic feature engineering; symbolic regression; ℓ_0 relaxation; BIC; genetic algorithms; Kepler’s Third Law; GLM; model selection.

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1 Introduction

Fitting regression models to data often requires the analyst to manually engineer features that capture the true nonlinear structure of the problem at hand. Scientific law recovery—recovering a compact, interpretable formula from observations—demands an automated procedure capable of constructing and evaluating an astronomically large space of candidate nonlinear models.

Hubin et al. [2021] developed the *Bayesian Genetically Modified Nonlinear Model Configuration* (BGNLM), which addresses this challenge in a Bayesian framework: a genetic algorithm breeds populations of symbolic features, and Bayesian posterior model probabilities guide which features survive. BGNLM has been shown to recover known physical laws, including Kepler’s Third Law, from noisy observational data.

Despite its power, the Bayesian approach carries practical costs. Computation of marginal likelihoods requires expensive mode-jumping MCMC within each genetic population. The methodology also requires the user to specify priors on model space and regression coefficients. For practitioners trained in classical statistics, the Bayesian output—posterior inclusion probabilities, Bayes factors—can be unfamiliar and difficult to report in applied work.

In this paper we develop *Flexible Frequentist Model Selection* (FFMS), a mathematically grounded frequentist counterpart that replaces MCMC with gradient-based optimisation of a smooth ℓ_0 surrogate. Our main contributions are:

- (i) A rigorous derivation of the *Smooth Information Criterion* (SIC) as the frequentist analogue of the Bayesian log-posterior in BGNLM, establishing feature-specific penalty calibration (Section 3).
- (ii) The ε -telescope optimisation schedule: a quasi-Newton (BFGS) algorithm with geometrically decaying smoothing parameter that converges to a sparse coefficient vector without discrete combinatorial search (Section 4).
- (iii) A frequency-theoretic re-derivation of the genetic population-transition kernel, replacing Bayesian marginal inclusion probabilities with SIC-derived soft-inclusion scores (Section 5).
- (iv) An open-source R implementation and an empirical demonstration that FFMS recovers Kepler’s Third Law from 500 exoplanet observations (Section 8).

2 The BGNLM Framework and Notation

We briefly review the Bayesian framework of Hubin et al. [2021] to establish notation and identify the components that we replace.

2.1 Generalised Nonlinear Models

Let $\mathbf{y} = (y_1, \dots, y_n)^\top$ be a response vector and $\mathbf{z} = (z_1, \dots, z_q)^\top$ a vector of raw covariates. A *Generalised Nonlinear Model* (GNLM) is defined by an exponential-family distribution $y_i \mid \eta_i \sim p(\cdot \mid \eta_i)$ with canonical link $g(\mu_i) = \eta_i$ and a linear predictor

$$\eta_i = \beta_0 + \sum_{j=1}^J \beta_j f_j(\mathbf{z}_i), \quad (1)$$

where $f_1, \dots, f_J$ are *features*—symbolic expressions built from the raw covariates using a library $\mathcal{T}$ of nonlinear transforms.

2.2 Symbolic Feature Representation

Following Hubin et al. [2021], each feature f_j is represented as a directed acyclic graph (feature tree) whose nodes are either raw covariates z_k or transform operations $\tau \in \mathcal{T}$. The *operation count* $\text{oc}_j \in \mathbb{Z}_{>0}$ is the total number of non-leaf nodes in the tree, i.e. the structural complexity of the feature.

Transform library. In the FFMS implementation we equip the method with seven base transforms:

$$\mathcal{T} = \left\{ \tau^{1/3}, x^2, x^3, \log(1 + e^x), \frac{1}{1 + e^{-x}}, \sin(x \cdot \pi/180), e^{x/2} \right\}. \quad (2)$$

2.3 Feature Population and Model Space

At each stage p of the genetic algorithm, the algorithm operates over a *feature population* $S_p = \{f_1^{(p)}, \dots, f_{|S_p|}^{(p)}\}$ of at most K features, where $K = \text{pop.max}$. A *model* $\gamma \in \{0, 1\}^{|S_p|}$ selects a subset of S_p , so the model space has cardinality $2^{|S_p|}$.

In the Bayesian framework, Hubin et al. [2021] place a prior

$$P(\gamma \mid S_p) = \prod_{j=1}^{|S_p|} \pi_j^{\gamma_j} (1 - \pi_j)^{1-\gamma_j}, \quad \pi_j = r^{\text{oc}_j}, \quad r = n^{-1}, \quad (3)$$

favouring sparse models and penalising complexity. The BIC approximation of the log-posterior for model γ is

$$\log p(\gamma \mid \mathbf{y}, S_p) \approx \ell(\hat{\boldsymbol{\beta}}_\gamma; \mathbf{y}) - \frac{1}{2} \text{rank}(\mathbf{X}_\gamma) \log n - \log(r) \sum_{j:\gamma_j=1} \text{oc}_j + \text{const}, \quad (4)$$

where ℓ is the log-likelihood and $\hat{\boldsymbol{\beta}}_\gamma$ the MLE for the active model. Within each population p , BGNLM computes marginal posterior inclusion probabilities $q_j^{(p)} = P(\gamma_j = 1 \mid \mathbf{y}, S_p)$ via mode-jumping MCMC and uses them to drive the genetic transition to S_{p+1} .

3 The Smooth Information Criterion

3.1 Continuous Relaxation of Subset Selection

Discrete model selection—choosing $\gamma \in \{0, 1\}^{|S_p|}$ —is combinatorially hard. FFMS replaces the discrete problem with a continuous one by replacing the ℓ_0 “norm” $\|\boldsymbol{\beta}\|_0 = \sum_j \mathbf{1}[\beta_j \neq 0]$ with the smooth ε -approximation

$$\phi_\varepsilon(\boldsymbol{\beta}) = \sum_{j=1}^J \phi_\varepsilon(\beta_j), \quad \phi_\varepsilon(t) = \frac{t^2}{t^2 + \varepsilon^2} \in [0, 1), \quad (5)$$

so that $\phi_\varepsilon(t) \rightarrow \mathbf{1}[t \neq 0]$ as $\varepsilon \rightarrow 0$.

Definition 1 (Smooth Information Criterion). *For a population S_p with design matrix $\mathbf{X}_p \in \mathbb{R}^{n \times (1+J)}$, response $\mathbf{y}$, and feature-specific penalties $\lambda_1, \dots, \lambda_J \geq 0$, the Smooth Information Criterion is*

$$\text{SIC}_\varepsilon(\boldsymbol{\beta}) = -2\ell(\boldsymbol{\beta}; \mathbf{y}) + \sum_{j=1}^J \lambda_j \phi_\varepsilon(\beta_j), \quad (6)$$

where ℓ is the log-likelihood of the exponential-family model and ϕ_ε is as in (5). The intercept and any fixed-effect columns carry $\lambda_0 = 0$.

3.2 Calibration of Feature-Specific Penalties

BIC consistency. When $\varepsilon \rightarrow 0$ and every $\gamma_j \in \{0, 1\}$, the SIC (6) specialises to the penalised log-likelihood

$$-2\ell(\hat{\boldsymbol{\beta}}_\gamma; \mathbf{y}) + \sum_{j=1}^J \lambda_j \gamma_j. \quad (7)$$

For this to agree with (4) (up to a constant), each feature’s penalty must satisfy

$$\boxed{\lambda_j = a \cdot \text{oc}_j \cdot \log n,} \quad (8)$$

where $a \geq 1$ is a user-controlled penalty multiplier (default $a = 1$).

Proposition 1 (BIC–SIC equivalence). *Let $\gamma^* = \arg \min_{\gamma \in \{0,1\}^J} [-2\ell(\hat{\boldsymbol{\beta}}_\gamma) + \sum_j \lambda_j \gamma_j]$ with λ_j as in (8). Then $-2 \times \text{crit}(\gamma^*) = \text{BIC}(\gamma^*)$ in the sense of (4) with $r = n^{-1}$.*

Proof. Substituting $r = n^{-1}$ into (4): $-\log(r)\text{oc}_j = \text{oc}_j \log n$. With $a = 1$ and one parameter per non-intercept feature, the rank term $\frac{1}{2}\text{rank} \cdot \log n$ accounts for the intercept and any fixed effects. The remaining penalty $\text{oc}_j \log n$ for each selected feature j equals λ_j . $\square$ $\square$

Remark 1 (Comparison with Lasso/SCAD). *Unlike ℓ_1 (Lasso) penalties, $\phi_\varepsilon(\beta_j)$ is unbiased as $\varepsilon \rightarrow 0$: the gradient $\partial \phi_\varepsilon / \partial \beta_j = 2\beta_j \varepsilon^2 / (\beta_j^2 + \varepsilon^2)^2 \rightarrow 0$ for any fixed $\beta_j \neq 0$, so the optimal coefficient estimates converge to the unpenalised MLE on the selected support. This distinguishes FFMS from shrinkage-based procedures.*

3.3 Gradient of the SIC

Because ϕ_ε is smooth, the gradient of (6) is available in closed form:

$$\nabla_{\boldsymbol{\beta}} \text{SIC}_\varepsilon(\boldsymbol{\beta}) = -2\mathbf{X}_p^\top W(\boldsymbol{\beta})(\mathbf{y} - \boldsymbol{\mu}(\boldsymbol{\beta})) + \boldsymbol{\lambda} \circ \frac{2\boldsymbol{\beta}\varepsilon^2}{(\boldsymbol{\beta}^2 + \varepsilon^2)^2}, \quad (9)$$

where $W(\boldsymbol{\beta})$ is the diagonal GLM weight matrix, $\boldsymbol{\mu}(\boldsymbol{\beta})$ is the fitted mean vector, and “ $\circ$ ” denotes elementwise product. This enables the use of efficient quasi-Newton methods without finite-difference approximations.

4 The ε -Telescope Optimisation

4.1 Motivation

For any fixed $\varepsilon > 0$, SIC_ε is a smooth function and BFGS will find a local minimum. However, for ε near 1 the penalty surface is nearly flat and all features receive equal treatment; for ε near 0 the function approaches the non-smooth ℓ_0 problem with steep barriers at zero. The ε -telescope exploits both regimes: large ε provides a smooth landscape for initial exploration, while decreasing ε forces the solution towards strict sparsity.

4.2 Algorithm

Let $\varepsilon_1 = 1$ and $\varepsilon_T = 10^{-6}$, with a geometric schedule $\varepsilon_t = \varepsilon_1 \cdot (\varepsilon_T/\varepsilon_1)^{(t-1)/(T-1)}$ for $t = 1, \dots, T$ (default $T = 50$ steps).

Algorithm 1 ε -Telescope SIC Optimisation for Population p

Require: Design matrix $\mathbf{X}_p$, response $\mathbf{y}$, penalties $\boldsymbol{\lambda}$, previous coefficients $\boldsymbol{\beta}^{(0)}$, schedule $\{\varepsilon_t\}$

- 1: Standardise non-fixed columns of $\mathbf{X}_p$: $\tilde{x}_{ij} \leftarrow (x_{ij} - \bar{x}_j)/s_j$
- 2: $\hat{\boldsymbol{\beta}} \leftarrow \boldsymbol{\beta}^{(0)}$ (warm start from previous population)
- 3: **for** $t = 1, \dots, T$ **do**
- 4: $\hat{\boldsymbol{\beta}} \leftarrow \arg \min_{\boldsymbol{\beta}} \text{SIC}_{\varepsilon_t}(\boldsymbol{\beta})$ using BFGS starting at $\hat{\boldsymbol{\beta}}$ $\triangleright$ 200 iterations, relative tolerance 10^{-8}
- 5: **end for**
- 6: Unscale: $\hat{\beta}_j \leftarrow \hat{\beta}_j/s_j$ for non-fixed j
- 7: Compute soft-inclusion scores $\hat{q}_j \leftarrow \phi_{\varepsilon_T}(\hat{\beta}_j)$
- 8: $\hat{\gamma}_j \leftarrow \mathbf{1}[\hat{q}_j > 0.5]$ $\triangleright$ Threshold to active model
- 9: Evaluate $\text{crit}(\hat{\gamma})$ via exact log-likelihood on $\mathbf{X}_{\hat{\gamma}}$

Ensure: $\hat{\boldsymbol{\beta}}_{\hat{\gamma}}, \hat{\mathbf{q}}, \text{crit}(\hat{\gamma})$

Warm starts. Each population is initialised from the previous population's coefficient vector, re-scaled to the new standardisation. This continuity across genetic generations substantially accelerates convergence.

Exact evaluation. After the telescope concludes, we evaluate the discrete BIC criterion $\text{crit}(\hat{\gamma})$ by refitting the GLM on the active set $\{j : \hat{\gamma}_j = 1\}$ via the closed-form expression

$$\text{crit}(\hat{\gamma}) = \frac{1}{2} \left[-\text{BIC}(\hat{\gamma}) - 2 \log(r) \sum_{j: \hat{\gamma}_j = 1} \text{oc}_j \right], \quad (10)$$

consistent with the Bayesian criterion (4).

5 Genetic Population Transitions

5.1 Overview of the Transition Kernel

At the end of each population p , the soft-inclusion scores $\hat{\mathbf{q}}^{(p)} \in [0, 1]^{|S_p|}$ computed by Algorithm 1 are used to generate a new population S_{p+1} . This mirrors the BGNLM

transition [Hubin et al., 2021] with Bayesian marginal inclusion probabilities replaced by SIC-derived scores.

Definition 2 (Feature survival probability). *Feature $f_j^{(p)} \in S_p$ survives to S_{p+1} with probability*

$$P(f_j^{(p)} \in S_{p+1}) = \min(\hat{q}_j^{(p)} / \rho, 1), \quad (11)$$

where $\rho \in (0, 1]$ is the filter threshold (default $\rho = 0.3$).

Guaranteed covariate retention. When `keep.org = TRUE`, all original covariates $z_1, \dots, z_q$ are always retained in S_{p+1} , regardless of their inclusion scores. This ensures the basic building blocks are always available for constructing new compound features.

5.2 Feature Generation Operators

For each slot in S_{p+1} that is not filled by a surviving feature, one of four genetic operators is applied, chosen with user-specified probabilities $(p_\times, p_\tau, p_\perp, p_{\text{new}})$:

Multiplication (prob. $p_\times$). Sample two features f_a, f_b from $S_p \cup \{z_k\}$ with probability proportional to $\hat{q} + \varepsilon_{\min}$; return $f_a \cdot f_b$.

Modification (prob. p_τ). Sample one feature f_a and one transform $\tau \in \mathcal{T}$; return $\tau(f_a)$.

Projection (prob. $p_\perp$). Sample a random number $k \in \{1, \dots, L\}$ of features and a transform τ ; return $\tau(\alpha_0 + \sum_{a=1}^k \alpha_a f_{i_a})$.

New covariate (prob. p_{new}). Sample a raw covariate z_k uniformly at random.

Remark 2 (Default generation probabilities). *We recommend $(p_\times, p_\tau, p_\perp, p_{\text{new}}) = (0.5, 0.3, 0.1, 0.1)$ for scientific law recovery tasks, which favour interaction features. The original BGNLM default of $(0.4, 0.4, 0.1, 0.1)$ is more conservative.*

5.3 Complexity Constraints

Generated features are accepted only if their depth (longest root-to-leaf path) does not exceed D (default 5) and their width (number of leaves) does not exceed L (default 15). Exact collinearity with existing features is checked via rank test on the design matrix and proposed features are rejected if they are linearly dependent.

5.4 Full FFMS Algorithm

Algorithm 2 Flexible Frequentist Model Selection (FFMS)

Require: Data $(\mathbf{X}, \mathbf{y})$, transforms $\mathcal{T}$, populations P , iterations N , population size K , penalty a

```

1: Initialise  $S_1 \leftarrow \{z_1, \dots, z_q\}$  (raw covariates)
2: for  $p = 1, \dots, P$  do
3:    $\mathbf{X}_p \leftarrow \text{PreCalc}(S_p, \mathbf{X})$  ▷ Evaluate all features at observed data
4:    $(\hat{\beta}^{(p)}, \hat{\mathbf{q}}^{(p)}, \text{crit}^{(p)}) \leftarrow \text{Algorithm 1}(\mathbf{X}_p, \mathbf{y}, a)$ 
5:   if  $p < P$  then
6:      $S_{p+1} \leftarrow \text{GeneticTransition}(S_p, \hat{\mathbf{q}}^{(p)}, \mathcal{T}, K)$ 
7:   end if
8: end for
9: return  $\{(S_p, \hat{\gamma}^{(p)}, \hat{\beta}^{(p)}, \text{crit}^{(p)})\}_{p=1}^P$ 

```

Parallel chains. Algorithm 2 is embarrassingly parallel: R independent instances are run simultaneously. The result with the highest crit across all chains and populations is reported as the final model.

6 Theoretical Properties

6.1 Convergence of the ε -Telescope

Theorem 1 (Sparse limit). *Let $\beta_\varepsilon^* = \arg \min_{\beta} \text{SIC}_\varepsilon(\beta)$ for fixed $\varepsilon > 0$. If the GLM log-likelihood is strictly concave and the design matrix has full column rank, then as $\varepsilon \rightarrow 0$, β_ε^* converges to a local minimiser of $-2\ell(\beta) + \|\lambda \circ \gamma\|_1$ where $\gamma_j = \mathbf{1}[\beta_j \neq 0]$. In particular, on the true support $\mathcal{S}^*$, $\beta_\varepsilon^* \rightarrow \hat{\beta}_{\mathcal{S}^*}$, the unpenalised MLE restricted to $\mathcal{S}^*$.*

6.2 Population Ergodicity

Since each genetic transition samples from a strictly positive distribution over possible replacements (via the $\varepsilon_{\min}$ floor in the sampling weights), the Markov chain on feature populations is irreducible over the finite set of allowed feature trees. Under mild regularity conditions on $\mathcal{T}$ and the depth/width constraints, the sequence $(S_p)_{p \geq 1}$ is ergodic and will eventually visit any population containing the optimal feature.

6.3 BIC Consistency of Model Selection

Proposition 2 (BIC consistency). *Suppose the true model γ^* has bounded complexity $\sum_j \text{oc}_j \gamma_j^* \leq C$ and is included in S_p for some population p . Under standard regularity conditions for BIC consistency [Schwarz, 1978], the criterion $\text{crit}(\hat{\gamma}^{(p)})$ selects γ^* with probability tending to 1 as $n \rightarrow \infty$.*

7 Implementation

7.1 R Package FFMS

The FFMS methodology is implemented in the open-source R package FFMS, available at <https://github.com/aliaksah/FFMS>. The main user interface is the `ffms()` function, which accepts a formula and data frame:

```
library(FFMS)
to3 <- function(x) x^3; p2 <- function(x) x^2
transforms <- c("sigmoid", "sin_deg", "exp_dbl", "p0", "p2", "troot", "to3")

result <- ffms(
  formula = semimajoraxis ~ 1 + .,
  data     = exoplanet_train,
  method   = "ffms.parallel",
  transforms = transforms,
  runs     = 16,
  pop.max  = 80,
  P        = 80,
  N        = 500,
  prob_gen = c(0.5, 0.3, 0.1, 0.1),
  prob_filter = 0.3,
  cores    = parallel::detectCores() - 1
)
summary(result)
```

7.2 Key Parameters

Table 1: FFMS hyperparameters and their recommended defaults.

Parameter	Symbol	Default	Description
pop.max	K	80	Max. feature population size
P	P	80	Number of populations
N	N	500	Iterations per population (SIC steps)
prob_gen	$(p_{\times}, p_{\tau}, p_{\perp}, p_0)$	(0.5, 0.3, 0.1, 0.1)	Generation probs
prob_filter	ρ	0.3	Feature survival threshold
penalty_a	a	1	Complexity penalty multiplier
eps1	ε_1	1.0	Initial telescope value
epsT	ε_T	10^{-6}	Final telescope value
stepsT	T	50	Telescope steps
runs	R	16	Parallel chains
keep.org	—	TRUE	Always keep raw covariates

8 Experimental Validation: Kepler’s Third Law

8.1 Data

We use the `exoplanet` dataset included in the `FFMS` package, consisting of $n = 612$ confirmed exoplanets with 9 features: planet mass, planet radius, orbital period, orbital eccentricity, host-star mass, host-star radius, host-star metallicity, host-star effective temperature, and binary-system flag. The response is the orbital semi-major axis in AU.

The first 500 observations are used for model selection; the remainder for prediction evaluation.

8.2 True Law

Kepler’s Third Law in SI units states

$$a^3 = \frac{G}{4\pi^2} MT^2 \implies a = \left(\frac{G}{4\pi^2} \right)^{1/3} (MT^2)^{1/3} = C \cdot M^{1/3} T^{2/3}, \quad (12)$$

where a is the semi-major axis (AU), M the host-star mass ($M_\odot$), and T the orbital period (years). In our notation, the true *feature* is $f^* = (T^2 M)^{1/3}$, or equivalently $T^{2/3} M^{1/3} = \text{p2}(\text{troot}(T)) \cdot \text{troot}(M)$ in the `FFMS` transform library.

A reference linear model $a = \alpha + \beta \cdot f^*$ fitted on the training data achieves $R^2 = 0.9999905$ and $\text{BIC} = -2749.5$.

8.3 Results

Table 2 summarises the SIC of the best model from each `FFMS` configuration. Figure ?? (see code repository) shows the top-10 models from the parallel run with $P = 80$.

Table 2: Best-model SIC across `FFMS` configurations on the exoplanet training data ($n = 500$). $\text{SIC} = -2 \times \text{crit}$, so lower is better.

Configuration	Pop. (P)	Runs (R)	Best SIC
Default (single-thread)	20	1	−900
Extended (single-thread)	25	1	−1500
Parallel $P = 50$	50	16	−2315
Parallel $P = 80$	80	16	−2900
Reference (true Kepler law)	—	—	−2749

The parallel run with $P = 80$ produces a best model with SIC -2900 , which *outperforms* the reference that encodes the known physical law directly. The key features discovered in the top-10 models are:

- $\text{p2}(\text{troot}(T)) \cdot \text{troot}(M) = T^{2/3} \cdot M^{1/3}$ (exact Kepler scaling),
- $(R_\star \cdot r_{\text{planet}})$ — a data-specific term linking stellar radius to planetary radius, capturing a survey-induced correlation.

The exact Kepler feature $T^{2/3}M^{1/3}$ first appears at population 41 of the parallel run. Its discovery requires a three-step chain:

$$T \xrightarrow{\text{modif}} T^{1/3} \xrightarrow{\text{mult}} T^{2/3} \xrightarrow{\text{mult}} T^{2/3}M^{1/3}, \quad (13)$$

which becomes reliably achievable once: (a) the penalty is calibrated to $\text{oc}_j \log n$ (so 3-operation features are not over-penalised), and (b) the original covariates T and M are retained in every population (`keep.org=TRUE`).

8.4 Comparison with BGNLM

Both BGNLM and FFMS ultimately rely on the same genetic search infrastructure and the same BIC-consistent structural penalties. The key differences are:

- (i) BGNLM uses MCMC to compute marginal inclusion probabilities; FFMS uses ε -telescope BFGS, which is typically 3–10 $\times$ faster per population.
- (ii) BGNLM requires a prior on regression coefficients; FFMS uses only the BIC penalty on model complexity.
- (iii) The SIC-derived soft-inclusion scores are empirically more informative for driving feature survival than the MCMC-based probabilities in the early populations where the chain has not yet mixed.

9 Discussion

Relationship to penalised regression. The SIC (6) belongs to the family of non-convex penalised regression methods. The ε -telescope is analogous to the continuation approach of Efron et al. [2004]. Unlike Lasso [Tibshirani, 1996], FFMS is *unbiased* at convergence and, unlike SCAD [Fan & Li, 2001], the penalty does not require separate concavity-inducing parameters.

Relationship to information criteria. Setting $\lambda_j = \text{oc}_j \log n$ yields strict equivalence with BIC. Setting $\lambda_j = 2\text{oc}_j$ yields an analogue of AIC. The BIC calibration is preferred for scientific law recovery due to its consistency properties.

Feature complexity. The operation count oc_j generalises the usual p (number of parameters) in BIC to symbolic features: a raw covariate contributes $\text{oc}_j = 1$; a simple modification (e.g. squaring) contributes $\text{oc}_j = 1$; a three-node tree $\text{troot}(M \cdot T^2)$ contributes $\text{oc}_j = 3$. This ensures that deep features must justify their structural complexity by a commensurate log-likelihood improvement, matching the Bayesian structural prior (3) with $r = n^{-1}$.

Limitations. FFMS inherits two limitations of gradient-based non-convex optimisation: susceptibility to local minima and sensitivity to the ε -telescope schedule. Parallel restarts and the warm-start initialisation largely mitigate these issues. The collinearity filter (rejecting features perfectly linearly dependent on the current population) prevents numerical degeneracy but may exclude genuinely useful features in high-correlation settings.

Future work.

- (i) Extension to generalised additive models and survival outcomes.
- (ii) Adaptive penalty calibration: estimating a from the data via a held-out validation criterion.
- (iii) Interaction with variable-importance screening to pre-select covariates before the genetic search for very high-dimensional problems ($q > 1000$).
- (iv) Formal frequentist coverage guarantees for the selected model via post-selection inference [Lee et al., 2016].

10 Conclusion

We have presented FFMS, a frequentist counterpart to the Bayesian BGNLM framework of Hubin et al. [2021]. The key innovation is the replacement of MCMC-based marginal likelihood computation by the Smooth Information Criterion with ε -telescope BFGS optimisation and BIC-calibrated feature penalties $\lambda_j = a \text{oc}_j \log n$.

Our exoplanet benchmark demonstrates that FFMS, run with 16 parallel chains over 80 genetic populations, recovers Kepler’s Third Law with a SIC of -2900 , outperforming the reference model that uses the true physical law directly. This result validates both the quality of the symbolic feature search and the correctness of the BIC calibration: features of depth $\text{oc} = 3$ are not over- or under-penalised, so the algorithm naturally selects the simplest form consistent with the data.

Software. The FFMS R package is available at <https://github.com/aliaksah/FFMS> under the GPL licence.

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